

IN3140 — Obligatorisk oppgave 3

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Task 1

1a)

Because the particle moves in a vertical movement it is parallel to the z-axis. So we have the height $h = y$. If we derivate the point we have the velocity $v = y' = \dot{y}$. And if we derivate the velocity we have the acceleration $a = v' = y'' = \ddot{y}$. We know that the langrangian for the particle is the potential energy minus the kinetic energy ($\mathcal{L} = K - P$).

The formula for the potential therefore is:

$$P = mgh = mgy$$

And the formula for the kinetic energy is:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{y}^2$$

And if we put it together:

$$L = K - P = \frac{1}{2}m\dot{y}^2 - mgy$$

From here to find the force (f) we must derivate L with respect to the time derivated to $\dot{q}$ which in this case equals $\dot{y}$ (like with a prismatic joint). And subtract the derivative of the langrangian with respect to q which in this case is y . So we have:

$$\frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial \dot{y}} = m\dot{y}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{d}{dt} \frac{\partial L}{\partial \dot{y}} = m\ddot{y}$$

$$\frac{\partial L}{\partial q} = \frac{\partial L}{\partial y} = -mg$$

$$f = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \frac{d}{dt} \frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = m\ddot{y} + mg$$

And if we rearrange the equation we get:

$$\underline{m\ddot{y} = f - mg}$$

1b)

From equation 9 in the task we have:

$$J = \begin{bmatrix} -s_1(L_2c_2 + L_3c_{23}) & -c_1(L_2s_2 + L_3s_{23}) & -c_1(L_3s_{23}) \\ c_1(L_2c_2 + L_3c_{23}) & -s_1(L_2s_2 + L_3s_{23}) & -s_1(L_3s_{23}) \\ 0 & L_2c_2 + L_3c_{23} & L_3c_{23} \\ 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix}$$

Finding the potential energy:

$$P = P_1 + P_2 = m_1gh_1 + m_2gh_2$$

- Because the first mass is evenly distributed, the mass center is found in the middle of link 1, so the height from the base of the robot to m_1 is half the length of link 1.
- Because the second mass is a point mass located at joint 3 (end of link 2), the height from the base of the robot to m_2 is defined as the z-coordinate in the matrix T_2^0 which is $L_1 + L_2s_2$ (the sign for L_2s_2 is changed from my first assignment because the configuration in this task is flipped 180 degrees from my configuration)

So the total potential energy is:

$$\begin{aligned} P &= m_1 9.81 \frac{1}{2} L_1 + m_2 9.81 (L_1 + L_2 s_2) \\ &= \underline{m_1 9.81 \frac{1}{2} L_1 + m_2 L_1 9.81 + m_2 L_2 s_2 9.81} \end{aligned}$$

Finding the kinetic energy:

$$K = K_1 + K_2 = \frac{1}{2} m_1 v_1^T v_1 + \frac{1}{2} \omega_1^T I_1 \omega_1 + \frac{1}{2} m_2 v_2^T v_2 + \frac{1}{2} \omega_2^T I_2 \omega_2$$

Starting with the linear potential energy (uses the 1. and 2. column of the jacobian in the assignment because we only use two links in this task):

$$v_1 = J_{v_1} \dot{q} = \begin{bmatrix} -s_1(L_2c_2 + L_3c_{23}) & -c_1(L_2s_2 + L_3s_{23}) \\ c_1(L_2c_2 + L_3c_{23}) & -s_1(L_2s_2 + L_3s_{23}) \\ 0 & L_2c_2 + L_3c_{23} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Sets all values related to θ_2 and θ_3 to 0:

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{v}_2 = \mathbf{J}_{\mathbf{v}_2} \dot{\mathbf{q}} = \begin{bmatrix} -s_1(L_2 c_2 + L_3 c_{23}) & -c_1(L_2 s_2 + L_3 s_{23}) \\ c_1(L_2 c_2 + L_3 c_{23}) & -s_1(L_2 s_2 + L_3 s_{23}) \\ 0 & L_2 c_2 + L_3 c_{23} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Sets all values related to θ_3 to 0:

$$= \begin{bmatrix} -s_1(L_2 c_2) & -c_1(L_2 s_2) \\ c_1(L_2 c_2) & -s_1(L_2 s_2) \\ 0 & L_2 c_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} -\dot{\theta}_1 s_1(L_2 c_2) - \dot{\theta}_2 c_1(L_2 s_2) \\ \dot{\theta}_1 c_1(L_2 c_2) - \dot{\theta}_2 s_1(L_2 s_2) \\ \dot{\theta}_2 L_2 c_2 \end{bmatrix}$$

Calculating $\mathbf{v}_1^2$ and $\mathbf{v}_2^2$:

$$\mathbf{v}_1^2 = 0$$

$$\mathbf{v}_2^2 = \mathbf{v}_2^T \mathbf{v}_2$$

$$= \begin{bmatrix} -\dot{\theta}_1 s_1 L_2 c_2 - \dot{\theta}_2 c_1 L_2 s_2 & \dot{\theta}_1 c_1 L_2 c_2 - \dot{\theta}_2 s_1 L_2 s_2 & \dot{\theta}_2 L_2 c_2 \end{bmatrix} \begin{bmatrix} -\dot{\theta}_1 s_1 L_2 c_2 - \dot{\theta}_2 c_1 L_2 s_2 \\ \dot{\theta}_1 c_1 L_2 c_2 - \dot{\theta}_2 s_1 L_2 s_2 \\ \dot{\theta}_2 L_2 c_2 \end{bmatrix}$$

$$= (-\dot{\theta}_1 s_1 L_2 c_2)^2 + \cancel{2\dot{\theta}_1 s_1 L_2 c_2 \dot{\theta}_2 c_1 L_2 s_2} + (-\dot{\theta}_2 c_1 L_2 s_2)^2 \\ + (\dot{\theta}_1 c_1 L_2 c_2)^2 - \cancel{2\dot{\theta}_1 c_1 L_2 c_2 \dot{\theta}_2 s_1 L_2 s_2} + (-\dot{\theta}_2 s_1 L_2 s_2)^2 \\ + (\dot{\theta}_2 L_2 c_2)^2$$

$$= (\dot{\theta}_1 s_1 L_2 c_2)^2 + (\dot{\theta}_1 c_1 L_2 c_2)^2 + (\dot{\theta}_2 c_1 L_2 s_2)^2 + (\dot{\theta}_2 s_1 L_2 s_2)^2 + (\dot{\theta}_2 L_2 c_2)^2 \\ = \cancel{(s_1^2 + c_1^2)}(\dot{\theta}_1 L_2 c_2)^2 + \cancel{(s_1^2 + c_1^2)}(\dot{\theta}_2 L_2 s_2)^2 + (\dot{\theta}_2 L_2 c_2)^2 \\ = (\dot{\theta}_1 L_2 c_2)^2 + (\dot{\theta}_2 L_2 s_2)^2 + (\dot{\theta}_2 L_2 c_2)^2 \\ = (\dot{\theta}_1 L_2 c_2)^2 + \cancel{(s_2^2 + c_2^2)}(\dot{\theta}_2 L_2)^2 \\ = \underline{\dot{\theta}_1^2 L_2^2 c_2^2 + \dot{\theta}_2^2 L_2^2}$$

Then calculates the angular potential energy:

$$\boldsymbol{\omega}_1 = \mathbf{J}_{\boldsymbol{\omega}_1} \dot{\mathbf{q}} = \begin{bmatrix} 0 & s_1 \\ 0 & -c_1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Sets the second column to 0:

$$\boldsymbol{\omega}_1 = \mathbf{J}_{\boldsymbol{\omega}_1} \dot{\mathbf{q}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$$\boldsymbol{\omega}_2 = \mathbf{J}_{\boldsymbol{\omega}_2} \dot{\mathbf{q}} = \begin{bmatrix} 0 & s_1 \\ 0 & -c_1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \dot{\theta}_2 s_1 \\ -\dot{\theta}_2 c_1 \\ \dot{\theta}_1 \end{bmatrix}$$

Calculating the Inertia from inertia tensors:

$$I_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{m_1 r_1^2}{2} \end{bmatrix} \quad I_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$J_1 = R_{z,\theta_1} I_1 R_{z,\theta_1}^T = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{m_1 r_1^2}{2} \end{bmatrix} \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{m_1 r_1^2}{2} \end{bmatrix}$$

$$J_2 = R_{x,\theta_2} I_2 R_{x,\theta_2}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_2 & -s_2 \\ 0 & s_2 & c_2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_2 & s_2 \\ 0 & -s_2 & c_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Total kinetic energy:

$$\begin{aligned} K &= K_1 + K_2 \\ K_1 &= \frac{1}{2} m_1 v_1^T v_1 + \frac{1}{2} \omega_1^T J_1 \omega_1 \\ &= \frac{1}{2} m_1 * 0 + \frac{1}{2} \begin{bmatrix} 0 & 0 & \dot{\theta}_1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{m_1 r_1^2}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 0 & 0 & \dot{\theta}_1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \frac{m_1 r_1^2}{2} \end{bmatrix} = \frac{1}{2} \dot{\theta}_1^2 \frac{m_1 r_1^2}{2} = \frac{m_1 \dot{\theta}_1^2 r_1^2}{4} \\ K_2 &= \frac{1}{2} m_2 v_2^T v_2 + \frac{1}{2} \omega_2^T J_2 \omega_2 \\ &= \frac{1}{2} m_2 (\dot{\theta}_1^2 L_2^2 c_2^2 + \dot{\theta}_2^2 L_2^2) + \frac{1}{2} \begin{bmatrix} \dot{\theta}_2 s_1 & -\dot{\theta}_2 c_1 & \dot{\theta}_1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_2 s_1 \\ -\dot{\theta}_2 c_1 \\ \dot{\theta}_1 \end{bmatrix} \\ &= \frac{1}{2} m_2 (\dot{\theta}_1^2 L_2^2 c_2^2 + \dot{\theta}_2^2 L_2^2) + \frac{1}{2} * 0 = \frac{m_2 \dot{\theta}_1^2 L_2^2 c_2^2 + m_2 \dot{\theta}_2^2 L_2^2}{2} \\ K &= \frac{m_1 \dot{\theta}_1^2 r_1^2}{4} + \frac{m_2 \dot{\theta}_1^2 L_2^2 c_2^2 + m_2 \dot{\theta}_2^2 L_2^2}{2} \end{aligned}$$

Now we have every part of the equation and can compute the Langrange:

$$\begin{aligned} \mathcal{L} &= K - P = (K_1 + K_2) - (P_1 + P_2) \\ &= \left(\frac{m_1 \dot{\theta}_1^2 r_1^2}{4} + \frac{m_2 \dot{\theta}_1^2 L_2^2 c_2^2 + m_2 \dot{\theta}_2^2 L_2^2}{2} \right) - (m_1 9.81 \frac{1}{2} L_1 + m_2 L_1 9.81 + m_2 L_2 s_2 9.81) \\ \mathcal{L} &= \frac{m_1 \dot{\theta}_1^2 r_1^2}{4} + \frac{m_2 \dot{\theta}_1^2 L_2^2 c_2^2}{2} + \frac{m_2 \dot{\theta}_2^2 L_2^2}{2} - m_1 9.81 \frac{1}{2} L_1 - m_2 L_1 9.81 - m_2 L_2 s_2 9.81 \end{aligned}$$

1c)

$$\tau_i = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i}$$

Because both joints are revoloute:

$$\tau_i = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) - \frac{\partial \mathcal{L}}{\partial \theta_i}$$

Starts by calculating τ_1 :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} &= \frac{m_1 \dot{\theta}_1 r_1^2}{2} + m_2 \dot{\theta}_1 L_2^2 c_2^2 \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} \right) &= \frac{m_1 r_1^2 \ddot{\theta}_1}{2} + m_2 L_2^2 c_2^2 \ddot{\theta}_1 - 2m_2 L_2^2 \dot{\theta}_1 \dot{\theta}_2 c_2 s_2 \\ \frac{\partial \mathcal{L}}{\partial \theta_1} &= 0 \\ \tau_1 &= \frac{m_1 r_1^2 \ddot{\theta}_1}{2} + m_2 L_2^2 c_2^2 \ddot{\theta}_1 - 2m_2 L_2^2 \dot{\theta}_1 \dot{\theta}_2 c_2 s_2 \end{aligned}$$

Calculates τ_2 :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} &= m_2 \dot{\theta}_2 L_2^2 \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} \right) &= m_2 \ddot{\theta}_2 L_2^2 \\ \frac{\partial \mathcal{L}}{\partial \theta_2} &= -m_2 \dot{\theta}_1^2 L_2^2 c_2 s_2 - m_2 L_2 c_2 9.81 \\ \tau_2 &= m_2 \ddot{\theta}_2 L_2^2 + m_2 \dot{\theta}_1^2 L_2^2 c_2 s_2 + m_2 L_2 c_2 9.81 \end{aligned}$$

The Euler-Langrange:

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} \frac{m_1 r_1^2 \ddot{\theta}_1}{2} + m_2 L_2^2 c_2^2 \ddot{\theta}_1 - 2m_2 L_2^2 \dot{\theta}_1 \dot{\theta}_2 c_2 s_2 \\ m_2 \ddot{\theta}_2 L_2^2 + m_2 \dot{\theta}_1^2 L_2^2 c_2 s_2 + m_2 L_2 c_2 9.81 \end{bmatrix}$$

Rewriting to the form $D(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau$:

$$\tau = \begin{bmatrix} \frac{m_1 r_1^2}{2} + m_2 L_2^2 c_2^2 & 0 \\ 0 & m_2 L_2^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} -m_2 L_2^2 c_2 s_2 \dot{\theta}_2 & -m_2 L_2^2 c_2 s_2 \dot{\theta}_1 \\ m_2 L_2^2 c_2 s_2 \dot{\theta}_1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ m_2 L_2 c_2 9.81 \end{bmatrix}$$

Task 2

2a)

Because the potential energy is defined by $\sum_{i=1}^n m_i g^T r_{ci}$ where $n = 3$ (because we have three links) the total potential energy can be found by using this expression to calculate the potential energy in each of the three masses and add them together. The mass is defined in the task, and so is g . I found r_{ci} by making a list of matrixes ($r1$, $r2$ and $r3$) which are the z-coordinates (the last column/translation) of the three matrixes H_1^0 , H_2^0 and H_3^0 found from assignment 1 and 2. In each of the links the mass is located in the middle of the link. Therefore in link 1, I divided the height by 2, in link 2 i divided all variables related to link 2 by 2, and the same for link 3. This represents the height of the mass because the z-coordinate of the baseframe of the robot (coordinatesystem $x_0y_0z_0$) points straight upwards, and becomes the height related to the base.

Code:

```
def pot():
    prod1 = mass[0]*g.T*r[0]
    prod2 = mass[1]*g.T*r[1]
    prod3 = mass[2]*g.T*r[2]
    return prod1+prod2+prod3

def main():
    print("Potential energy:")
    pretty_print(pot())
```

Output:

```
potential energy:
[5.9316165·L1 + 2.715408·L2·sin(θ2) + 0.689643·L3·sin(θ2 + θ3)]
```

2b)

To calculate the kinetic energy i used the formula in the assignment

$\frac{1}{2} \dot{\mathbf{q}}^T [\sum_{i=1}^n \mathbf{m}_i \mathbf{J}_{vi}(\mathbf{q})^T \mathbf{J}_{vi}(\mathbf{q}) + \mathbf{J}_{wi}(\mathbf{q})^T \mathbf{R}_i(\mathbf{q}) \mathbf{I}_i \mathbf{R}_i(\mathbf{q})^T \mathbf{J}_{wi}(\mathbf{q})] \dot{\mathbf{q}}$ Where $n = 3$ because there are three masses. The masses are given in the assignment, where $\mathbf{m}_i$ represent the mass of mass point i. $\mathbf{J}_{vi}(\mathbf{q})$ represents the first three rows in the jacobian, and $\mathbf{J}_{wi}(\mathbf{q})$ represents the last three rows of the jacobian. The R matrix is the rotational matrix for each mass related to the baseframe, where the first matrix was given in the task, and the others were calculated by matrix multiplication with a z-rotation matrix. To calculate each joint by it self I matrix multiplied with a diagonal matrix so that the other non-relevant columns was set to zero, and to changed all non-relevant values to zero by using the function 'subs'. To calculate the total kinetic energy i added the kinetic energy from each mass.

Code:

```
def kin():
    col1 = diag(1,0,0)
    col2 = diag(1,1,0)
    Jv1 = Jv.subs([(cos(theta2), 0),
                  (cos(theta3), 0),
                  (sin(theta2), 0),
                  (sin(theta3), 0),
                  (cos(theta2+theta3), 0),
                  (sin(theta2+theta3), 0),
                  (L2, 0),
                  (L3, 0)]))
    Jv2 = Jv.subs([(cos(theta3), 0),
                  (sin(theta3), 0),
                  (cos(theta2+theta3), 0),
                  (sin(theta2+theta3), 0),
                  (L3, 0)]))
    Jv3 = Jv
    Jw1 = Jw.subs([(cos(theta2), 0),
                  (cos(theta3), 0),
                  (sin(theta2), 0),
                  (sin(theta3), 0),
                  (cos(theta2+theta3), 0),
                  (sin(theta2+theta3), 0),
                  (L2, 0), (L3, 0)]))
    Jw2 = Jw.subs([(cos(theta3), 0),
                  (sin(theta3), 0),
                  (cos(theta2+theta3), 0),
                  (sin(theta2+theta3), 0),
                  (L3, 0)]))
    Jw3 = Jw
```


2c)

$$\tau = D(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q)$$

Can be written as:

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} d_{11}(q) & d_{12}(q) & d_{13}(q) \\ d_{21}(q) & d_{22}(q) & d_{23}(q) \\ d_{31}(q) & d_{32}(q) & d_{33}(q) \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \end{bmatrix} + \begin{bmatrix} c_{11}(q, \dot{q}) & c_{12}(q, \dot{q}) & c_{13}(q, \dot{q}) \\ c_{21}(q, \dot{q}) & c_{22}(q, \dot{q}) & c_{23}(q, \dot{q}) \\ c_{31}(q, \dot{q}) & c_{32}(q, \dot{q}) & c_{33}(q, \dot{q}) \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} + \begin{bmatrix} g_1(q) \\ g_2(q) \\ g_3(q) \end{bmatrix}$$

Where d_{ij} are elements of matrix D , c_{ij} are elements of matrix C , g_k are elements of matrix g , τ_k are elements of matrix τ , $\ddot{q}_k$ are elements of matrix $\ddot{q}$ and $\dot{q}$ are elements of matrix $\dot{q}$

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} d_{11}(q)\ddot{q}_1 + d_{12}(q)\ddot{q}_2 + d_{13}(q)\ddot{q}_3 \\ d_{21}(q)\ddot{q}_1 + d_{22}(q)\ddot{q}_2 + d_{23}(q)\ddot{q}_3 \\ d_{31}(q)\ddot{q}_1 + d_{32}(q)\ddot{q}_2 + d_{33}(q)\ddot{q}_3 \end{bmatrix} + \begin{bmatrix} c_{11}(q, \dot{q})\dot{q}_1 + c_{12}(q, \dot{q})\dot{q}_2 + c_{13}(q, \dot{q})\dot{q}_3 \\ c_{21}(q, \dot{q})\dot{q}_1 + c_{22}(q, \dot{q})\dot{q}_2 + c_{23}(q, \dot{q})\dot{q}_3 \\ c_{31}(q, \dot{q})\dot{q}_1 + c_{32}(q, \dot{q})\dot{q}_2 + c_{33}(q, \dot{q})\dot{q}_3 \end{bmatrix} + \begin{bmatrix} g_1(q) \\ g_2(q) \\ g_3(q) \end{bmatrix}$$

And if you only view one row which represents one mass (k):

$$\tau_k = (d_{k1}(q)\ddot{q}_1 + d_{k2}(q)\ddot{q}_2 + d_{k3}(q)\ddot{q}_3) + (c_{k1}(q, \dot{q})\dot{q}_1 + c_{k2}(q, \dot{q})\dot{q}_2 + c_{k3}(q, \dot{q})\dot{q}_3) + (g_k(q))$$

Then we can put up some summations for the $\ddot{q}_j$ and $\dot{q}_j$, where n would be 3 in our case because we have 3 masses

$$\tau_k = \sum_{j=1}^3 (d_{kj}(q)\ddot{q}_j) + \sum_{j=1}^3 (c_{kj}(q, \dot{q})\dot{q}_j) + (g_k(q))$$

Where j is the index of each column in D and C . And from the textbook (page 183) we have that:

$$c_{kj} = \sum_{i=1}^n c_{ij}(q)\dot{q}_i$$

So we can write the expression like this:

$$\tau_k = \sum_{j=1}^3 (d_{kj}(q)\ddot{q}_j) + \sum_{i=1}^3 \sum_{j=1}^3 (c_{ij}(q)\dot{q}_i\dot{q}_j) + (g_k(q))$$

Forces represented by...

- $D(q)$ - the inertia ($n \times n$), and multiplied with $\ddot{q}$ expresses the acceleration for the motion ($n \times 1$)
- $C(q, \dot{q})$ - the coriolis ($n \times n$) (centrifugal forces), and multiplied with $\dot{q}$ expresses the quadric velocity ($n \times 1$)
- $g(q)$ - the gravitational force ($n \times 1$)

2d)

The two remaining expressions are for $C(q, \dot{q})$:

$$c_{kj} = \sum_{i=1}^n c_{ij}(q) \dot{q}_i = \sum_{i=1}^n \frac{1}{2} \left(\frac{\partial d_{kj}}{\partial q_1} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k} \right) \dot{q}_i$$

And the for $g(q)$:

$$g(q) = (g_1(q), \dots, g_n(q))$$

To calculate the D matrix I filled out the variables in the formula above. And g is given by $[0 \ 0 \ 9.81]^T$

Code:

```
def cor(D):
    C = zeros(3,3)
    for k in range(3):
        for j in range(3):
            for i in range(3):
                C[k,j] = (1/2)*(diff(D[k,j],thetas[i])
                               +diff(D[k,i],thetas[j])
                               -diff(D[i,j],thetas[k]))*qdot[i]
    return C

main():
    print("g(q):")
    pretty_print(g)
    print("\nC(q,qdot):")
    C = cor(D)
    pretty_print(C)
```

Output (only for g because C was way too long):

```
g(q):
[
  0
]
[
  0
]
[
  9.81
]
```

2e)

To assemble each τ_i I multiplied $D(q)(3 \times 3)$ with $\ddot{q}(3 \times 1)$ and $C(q, \dot{q})(3 \times 3)$ with $\dot{q}(3 \times 1)$. And then added all the parts together.

Code:

```
def tau(D,C,g):
    tau = zeros(3,1)
    Dpart = D*qdotdote
    Cpart = C*qdot
    return Dpart+Cpart+g

def main():
    t = tau(D,C,g)
    pretty_print(t)
```